



A review of edge computing reference architectures and a new global edge proposal

Inés Sittón-Candanedo ^{a,*}, Ricardo S. Alonso ^a, Juan M. Corchado ^{a,b,c,d},
Sara Rodríguez-González ^{a,*}, Roberto Casado-Vara ^a

^a Bisite Research Group, University of Salamanca, Salamanca, Spain

^b Air Institute, IoT Digital Innovation Hub, Salamanca, Spain

^c Department of Electronics, Information and Communication, Faculty of Engineering, Osaka Institute of Technology, Osaka, Japan

^d Pusat Komputeran dan Informatik, Universiti Malaysia Kelantan, Kelantan, Malaysia

HIGHLIGHTS

- This article is a review of the Edge Computing technology and its reference architectures.
- It presented a proposal for a tiered architecture with a modular approach that allows to manage the complexity of solutions.
- The main contributions of the proposed architecture reside in the security and privacy provided by blockchain technologies.
- The proposed reference architecture is tested by building an IoT platform in a smart agroindustry scenario.

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ABSTRACT

Edge Computing represents the activities of IoT (Internet of Things) devices at the border or limit of the network connected to the remote Cloud. The latest research in this field has intended to demonstrate that Edge Computing architectures are the optimal solution to minimising latency, improving privacy and reducing bandwidth costs in IoT-based scenarios. This article is a review of the Edge Computing technology and its reference architectures proposed by the Edge Computing Consortium, Intel-SAP, the FAR-Edge Project and the Industrial Internet Consortium for Industry 4.0. Moreover, this article presents a proposal for a tiered architecture with a modular approach that allows to manage the complexity of solutions not only for Industry 4.0 environments but also for other scenarios such as smart cities, smart energy, healthcare or precision agrotechnology. The main contributions of the proposed architecture reside in the security and privacy provided by blockchain technologies. Finally, the proposed reference architecture is tested by building an IoT platform in a smart agroindustry scenario to reduce bandwidth costs between the Edge and the Cloud.

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1. Introduction

The use of the Internet of Things (IoT), mobile services and applications in complex scenarios such as Smart Cities and Industry 4.0 has generated new challenges for Cloud Computing (CC) oriented solutions [1,2]. Traditional Cloud Computing follows a centralised scheme where computing and storage are deployed in a remote data centre [3]. However, this approach experiences significant limitations when dealing with these emerging technologies that require real-time response and reduced latency. In this regard, recent researches propose the use of the Edge

Computing paradigm as a means of improving Cloud Computing capabilities [4–6].

In recent years, Edge Computing has attracted the attention of enterprises, academics, industries and researchers. The first mobile Edge Computing platform, named RACS (*Radio Applications Cloud Server*), was presented by Nokia and IBM in 2013 [7,8]. However, the origins of Edge Computing go back as far as 1990 when Content Delivery Networks (CDNs) were proposed as a means of improving web performance. The Cisco Global Cloud Index estimated that by 2019 data produced by humans and machines will reach 500 zettabytes, and that by 2020 there will be 50 billion IoT devices connected to the Internet [9]. Edge Computing is a potential solution to this problem, a technology that optimises computing processes. Edge Computing does not process data directly in the Cloud, instead it preprocesses the data

* Corresponding authors.

E-mail addresses: isittonc@usal.es (I. Sittón-Candanedo), srg@usal.es (S. Rodríguez-González).

collected by IoT Edge devices before sending them to the remote centralised or distributed servers deployed in the Cloud [10,11].

It is denominated as “Edge” computing because information processing no longer takes place only in centralised or distributed nodes (core), but is performed also in the other extreme (Edges), moving centralised computing processes away from the centre. All the information produced by IoT devices is processed in the *extreme*, thus releasing computational load from centralised servers, avoiding network traffic overload, and reducing the response time required by new IoT applications [12]. The benefit of this type of distributed computing approach has been demonstrated in, for example, smart homes, healthcare, smart grid, smart cities, smart logistics and environment monitoring [1,13–15] or also in the use of *Cloudlets* (small Clouds at the edge) used to download the computing overload of wearable devices [16]; this helps save between 30% and 40% of energy [17].

Currently, there are several lines of research aimed at establishing standards for the development of concepts, architectures or processes implemented in Edge Computing solutions. In this regard, important independent organisations and entities have proposed different specifications, i.e.: communication protocols, security – data protection – as well as reference architectures, specifically for industrial environments [18–21]. All those specifications have contributed to some aspects of Industry 4.0 applications and environments. However, to the best of the authors’ knowledge, there has not yet been a standardised proposal that would focus on providing security to applications and environments, such as the inclusion of Distributed Ledger Technologies (DLT) or a blockchain based layer for the implementation of *smart contracts* between entities [22,23]. Hence, this work proposes a new Edge Computing reference architecture for the design of applications for Industry 4.0 and other scenarios, taking into consideration all these requirements.

This article is structured as follows: Section 2 reviews research in the field of Edge Computing; Section 3 presents the main reference architectures and describes some use cases based on the ISO/IEC/IEEE 2010:2011 standard proposed by four groups of major companies whose objective is to adopt Edge Computing as a means of strengthening Cloud-based solutions [24]; Section 4 proposes the new Global Edge Computing Architecture which is applied to an agroindustry use case in Section 5. Finally, the conclusions and future lines of research are presented in Section 6.

2. Edge Computing

In the *post-cloud* era, Edge Computing has emerged as a new paradigm in which part of the data is processed at the *Edge* of the network. The history of the Edge Computing (EC) paradigm was described in the work “On multi-access Edge Computing: a survey of the emerging 5G network Edge Cloud architecture and orchestration” [4]. This section presents a review of the basic elements of Edge Computing and describes the close relationship between IoT networks and Edge Computing.

To several authors, Edge Computing represents a group of computers, devices and network resources which generate and collect data and send it to remote centres in the Cloud. The works of these authors show Edge Computing from different perspectives, in terms of architectures, capabilities, software technologies or features [5,25,26].

A new group of industry specifications was proposed in 2014 with the objective of creating the European Telecommunications Standards Institute (ETSI). This institute seeks to establish Mobile Edge Computing (MEC) standards and specifications for industries, and has the support of a consortium formed by several organisations, including Huawei, IBM, Intel, Nokia Networks,

Table 1
Edge Computing definitions.

Authors	Definition
Garcia et al. [12]	Edge Computing refers to a process where open platforms converge the core capabilities of networks, computing, storage and applications and provide intelligent services at the network edge – near the source of the objects or data – to meet the critical requirements of agile connection, real-time services, data optimisation, application intelligence, security and privacy protection of industry digitisation.
Shi et al. [10]	Edge Computing refers to the enabling technologies that allow to perform computation at the network edge so that computing happens near data sources. In Edge Computing, the end device not only consumes data but also produces data.
Satyanarayanan [8]	Edge Computing is a new paradigm in which substantial computing and storage resources, also referred to as <i>Cloudlets</i> , <i>micro datacentres</i> or <i>fog nodes</i> , are placed at the Internet edge in close proximity of mobile devices or sensors.
Edge Computing Consortium, Alliance of Industrial Internet [20]	Edge Computing is a distributed open platform at the network edge, close to the things or data sources, integrating the capabilities of networks, storage, and applications. By delivering edge intelligence services, Edge Computing meets the key requirements of industry digitisation for agile connectivity, real-time services, data optimisation, application intelligence, security and privacy protection.
Tseng et al. [21]	Edge Computing, simply known as <i>Edge</i> , brings processing close to the data source, and it does not need to be sent to a remote Cloud or other centralised systems for processing. By eliminating the distance and time it takes to send data to centralised sources, the speed and performance of data transmission improves, as well as the devices and applications on the Edge.
Linthicum [29]	Edge Computing or simply <i>Edge</i> eliminates the need to process data in a remote Cloud or other centralised system by processing data close to its source. Data transmission is faster and the performance of devices and applications on the Edge improves.
Zhang et al. [30]	Edge Computing is a novel computing model that places computing resources and storage at the edge of the network, closer to the end user. It provides intelligent services by collaborating with Cloud Computing.

NTT DoCoMo and Vodafone [27]. In Bilal et al. [28] the authors’ highlight that the main objective of Edge Computing is to bring the resources and services of the Cloud to the Edge of the network, operating as an intermediate layer between the end user and the data centres in the Cloud, with a minimum delay in response time. Open Edge Computing (OEC) is a 2015 initiative undertaken by Vodafone, Intel and Huawei, in association with the Carnegie Mellon University, whose aim is to promote Edge Computing solutions. Furthermore, Nokia integrated an Edge Computing platform into its base station [7].

State-of-the-art researches have provided different Edge Computing definitions, Table 1 lists several key definitions. All the definitions found in Table 1 include the following three terms: *open platform*, *enabling technologies* and *computer resources*. However, in our view, the most accurate definition is the one established by the Edge Computing Consortium because it contains elements that organisations currently demand: *secure and private services that are capable of processing large volumes of data in real time*.

Table 2 compares the characteristics of Edge Computing with the characteristics of other technologies. It demonstrates how Edge Computing improves the performance of IoT: its distributed structure makes it possible to reduce network traffic; transmission latency between edge nodes, the cloud and end users; and

Table 2
Comparison of the characteristics of EC with those of other technologies.

Parameters	Cloud Computing	Fog Computing	Edge Computing
Advantages	Scalable, big data processing, unlimited computational processing.	User defined security, low latency.	Real time responses, very low latency, Edge can work without cloud or fog.
Disadvantages	High latency, slow response time, no offline-mode, lack of security.	Limited storage, Fog needs more cloud links than Edge Computing to move data from the physical to the digital layer.	Limited storage, Interconnected through proprietary networks, high power consumption.

finally the real-time response of applications and IoT devices compared to Cloud and Fog computing [31].

2.1. Edge Computing and IoT

Ganz et al. [32] view the Internet of Things (IoT) as the interaction and communication between devices that generate and exchange data with *things* or objects in the real world. Razzaque et al. [33] mention the main characteristics of IoT: heterogeneous devices and networks, large volume of events, and data generated by objects. In the authors' opinion these characteristics make the development of IoT-oriented applications and services very challenging. IoT devices have laid the foundations for the digitisation of the physical world thanks to their ability to transmit data that can be processed and analysed by machines, transforming them into valuable information used to create intelligent health, transportation, energy and other services [34].

The requirements for the development of applications in IoT and Cloud scenarios are becoming increasingly difficult to meet. The large volume of data generated by IoT sensors must be analysed to extract the information or statistics that the organisation is looking for, and this task requires a previous stage with an extremely high network bandwidth [35]. In this respect, previous researches have proposed the use of Edge Computing to find solutions that comply with the requirements of IoT and can use its potential and capabilities to the fullest. Brogi and Forti [36] explained that in Edge Computing the set of data generated by different IoT objects can be processed at the edge of the network instead of transmitting it to a remote centralised or distributed Cloud platform. This enables faster services and higher response quality compared to Cloud Computing. According to Lin et al. [37], architectures based on Edge Computing are the most suitable for integration with IoT, providing secure and efficient services to end users.

Security is challenging to achieve but it is an essential feature in any IoT scenario. In this regard, Khan and Salah [38] proposed the use of blockchain (BC) technologies as tools capable of dealing with the security issues experienced by the IoT. Jing et al. [35] addressed major challenges in the field of IoT security. Finally, in several studies [22,39,40], authors researched the advantages of integrating blockchain and IoT and the challenges that this entails. Even though blockchain has originally been created as a tool for cryptocurrencies, the use of a BC in decentralised applications does not necessarily have to involve the development of a cryptocurrency. As its name indicates, BC is a chain of time-stamped blocks (each one includes date and time) that are connected with each other by a cryptographic hash [41,42].

BC relies on the mining process to preserve the value and integrity of the data. As a result, a new block of data can only be added to the chain when a *miner* obtains the hash value that connects the last block with a new one. To complete the operation, it is necessary that the rest of miners across the network validate the result by means of a *consensus* protocol [43].

3. Edge Computing architectures

This section presents a review of reference architectures (RA) developed for Edge Computing. Recent research has focused on looking into the ability of the Edge Computing Paradigm to lessen the limitations of Cloud-centric architectures. In this respect, Edge Computing architectures are capable of shifting a portion of the computing capacity that is performed from the Cloud to the nodes located at the Edge of the network [44]. By shifting the processing capacity to the nodes, Edge architectures offer the following advantages [45,46]:

- Save bandwidth and storage resources; to filter noise data streams that come from different data sources are processed by the nodes before sending the data to the Cloud.
- Proximity and low latency are achieved by processing information close to its source of origin.
- Enhanced scalability is achieved through decentralised storage and processing.
- The nodes of Edge architectures provide each node of the network with isolation and privacy.

A Reference Architecture is a document or a set of documents that establish standards for the design of architectures and the integration of IT products and services. A reference architecture incorporates industry-accepted best practices, usually providing guidelines for the method of delivery or specific optimal technologies. RAs help project managers, software developers, enterprise designers, and IT managers to collaborate and communicate effectively in the implementation of a project. A reference architecture gives answers to frequently asked questions and any doubts that may arise [47]. Consequently, by providing a proven set of best practices and solutions, it helps teams avoid the errors and delays that would have occurred otherwise. Once they are designed, most reference architectures are tested extensively and validated before being documented by expert engineers. Since a reference architecture is normally implemented in many different situations, on the basis of those real experiences experts make adjustments to later stages. As a result of those development iterations companies benefit from the accumulated wisdom.

Reference Architectures are not complexly packaged and in many cases adopt a descriptive approach. This means they are flexible, allowing customers to create a solution that includes components from their preferred IT vendors. Some of the reference architectures that have been designed for Edge Computing are based on the ISO/IEC/IEEE 42010:2011 standard [24]. This standard outlines the requirements for the development of systems and software architectures through a convention, a common terminology and the best practices of Design and Description of Enterprise Architecture and it is based on a conceptual model or metamodel of Architecture Description (AD) [48].

3.1. FAR-Edge RA

The FAR-Edge RA has been developed as part of the H2020 FAR-Edge project, as a means of overcoming the challenge that industries face in adopting decentralised automation architectures [18]. This reference architecture is a conceptual framework for the design and implementation of the FAR-Edge project platform based on Edge Computing and ledger tier (DLT – *Distributed*

	Functional Components			Structural Components
	Automation	Analytics	Simulation	
Enterprise ecosystem	Applications			CLOUD
	Cloud Services			
	Ledger Services			LEDGER
Plant	Edge processes			EDGE
	Smart Objects			FIELD
	Connected Devices			
	Things, people & Environments			

Fig. 1. FAR-Edge RA. Based on the work of Isaja et al. [44].

Ledger Technologies – currently Blockchain). Isaja et al. [44] explained the two principal concepts of FAR-Edge: *scopes* and *tiers*. *Scopes* refer to elements in a plant or its ecosystems such as machinery, field devices, workstation, SCADA (Supervisory Control And Data Acquisition), MES (Manufacturing Execution System) and ERP (Enterprise Resource Planning) Systems. Likewise, *tiers* provide information about system components and the relationship between them. The classification of tiers is similar to the one described in OpenFog RA, the difference between the two is that Far-Edge RA is only oriented at the industry [49].

3.1.1. Components

Fig. 1 has been created on the basis of the work of Isaja et al. [44], it describes the main components of this architecture, such as its functional domains and tiered structure. The functional and structural components are described as follows:

- **Functional.** The architecture is formed by three functional domains: (i) *Automation Domain*: it is responsible for the automated control and configuration of physical processes. Automated configuration refers to the technology used for monitoring, customising and adjusting the critical processes of organisations, such as the production process of a factory. This technology controls the analytic and simulation domains. (ii) *Analytic Domain*: it contains the tools for the collection and processing of data from the perspective of business intelligence. The analytic domain uses a high bandwidth as a communication channel through which the bulk of information is transferred. The Analytic Domain of the FAR-Edge architecture and the Information Domain of the Industrial Internet Consortium Reference Architecture (IIC-RA) mostly coincide with each other and have only minimal differences, for example: the IIC-RA receives data directly from the field with the intervention of additional control functionalities. (iii) *Simulation domain*: this domain contains resources for the simulation of the behaviour of physical processes, which are performed to analyse different *what/if* scenarios, optimising and obtaining results without affecting operational activities in any way. In the simulation domain, objects from the real environment must be synchronised with the digital simulation models that represent them.
- **Structural.** It refers to components and technical levels which form the architecture. These tiers are described in the following paragraphs:

- **Field tier:** is the lowest layer of the architecture and it is formed by different types of devices, ranging from a smart machine to a sensor or an actuator. These devices are called *Edge nodes*.

- **Edge tier:** is the software designed to run on several Edges nodes. Moreover, this tier contains the *Edge gateways* (EGs). EGs are computing devices that operate as gateways between the field tier and the digital world. The main function of EGs is to execute processes and analyse data in real time.
- **Ledger tier:** it does not contain any type of physical component. It is formed by services based on a chain of distributed blocks (Blockchain) and its function is to evaluate and approve each service call of the Edge nodes that need to be modified before it comes into effect. By means of *smart contracts* [40], the ledger tier regulates the architecture's functional components, i.e.: automation and analysis.
- **Cloud tier:** consists of Cloud Servers (CSs) that host the software responsible for planning, monitoring and managing the resources of the company. In resume, the Cloud tier deals with the logical execution of the functional components of the architecture.

The FAR-Edge reference architecture is notable for having implemented Edge Computing while guaranteeing information security through blockchain. By including automation and analysis through smart contracts, this reference architecture can be used as a reference for automation use cases, analysis and digital simulation in industrial environments [44].

3.1.2. Application scenario: Production plant

The implementation of the architecture in a production plant is one of the use cases proposed by the developers of the FAR-EDGE project. In this factory, the finished products are stacked on pallets that are transported by a single belt to a warehouse. Although each pallet contains only products of the same type, the belt transports different pallets that come from parallel assembly lines, therefore, the sequence of the products in the conveyor cannot be predicted. Once in the warehouse, pallets are sent to several docks for distribution, temporary storage or discarded as a defective product. The logic model must take into account the product type, its configuration, capacity and status. Currently, the PLC system coincides with the exit bay and its functions are to: dispatch the products; assign an ID to the product and to the pallet and takes into account the corresponding production line. The implementation of the FAR-Edge RA allows through its Ledger services the decentralised configuration and orchestration to synchronise the digital model of the plant with the real world. The Ledger level is in charge of granting permission for: the addition of new bays; the disconnection of existing bays at any time; the request of new products when the processed ones exceed the incoming flow. Blockchain is used in FAR-EDGE to implement Ledger services, avoiding possible breaches or scalability bottlenecks [44].

3.2. INTEL-SAP RA

INTEL and SAP have developed an Edge Computing RA. The main objective of this architecture is to allow for the fast and scalable development of IoT projects. Fig. 2 shows the functional blocks used by this RA. Below is a summary of those blocks (Intel-SAP joint reference architecture) [19]:

- **Edge Endpoint:** The IoT-Edge modules of the architecture receive messages sent by authorised devices and send messages from the gateway to the server in the Cloud. The INTEL-SAP RA uses persistence and streaming analysis services for local storage of data generated by sensors or IoT devices. It performs the analysis of IoT data flows in search of exceptions and patterns that allow for the creation of events or alerts according to the type of use case.

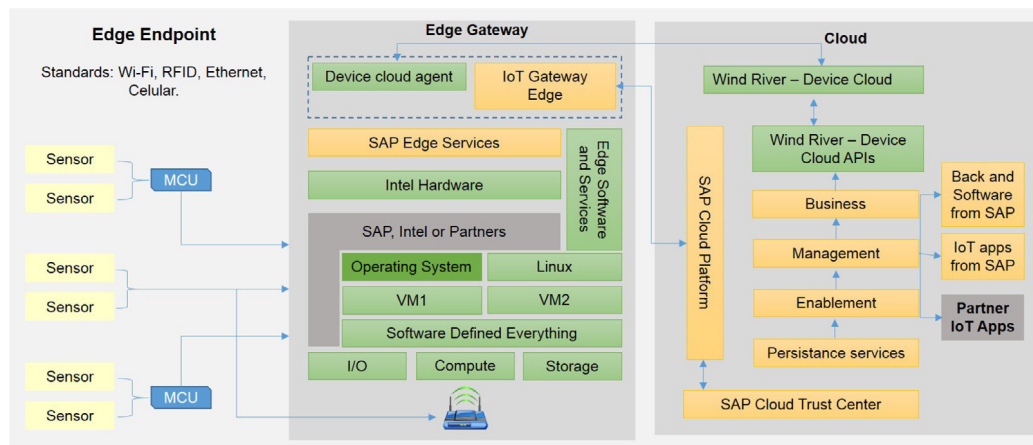


Fig. 2. INTEL-SAP RA. Based on work of [19].

- **Edge Gateway:** In this architecture its gateway is authenticated through the device certificate and establishes a data path to the Cloud Computing provider. Here, the incorporation of the device to establish the control and data path flows is completed. The device turns on and it communicates with an agent in the Intel Cloud. This agent provides the IP address and redirects the device to the SAP Cloud Trust Center. The applications in the gateway are fed with the data coming from the connected sensors and these are sent through protocols supported by the architecture.
- **Cloud:** In this block, the SAP Cloud Trust Center is responsible for verifying the Intel EPID signature and ownership of the device and then registering it. It generates a certificate of authenticity and keeps the list of authorised devices up to date. The connectivity path between the device and the Cloud is established. The generated certificates are sent for the configuration of publications–subscriptions at the gateway. Once the data has been sent to the Cloud services, it is distributed to different applications for its processing and it can be sent to back-end applications for use in other organisational processes.

According to its creators, this reference architecture groups together: the performance of the Intel IoT platform, the capabilities of SAP Leonardo IoT and SAP Cloud Platform to securely incorporate and manage a device network, collect and manage machine data and perform analysis [19].

3.2.1. Application scenario: Connected assets and logistics

Connected assets and the logistics use case demonstrate the value of applying portions of the joint reference architecture to a real-world example. This use case illustrates the simplified view of the combined physical and digital world linked to business processes. Although this use case focuses on an IoT deployment of connected assets in a discrete manufacturing environment, the joint reference architecture can be applied to other sectors, such as retail, industry, energy, bio-tech, pharmacy, chemistry, or electronics, where data from physical machinery or devices is generated and can be combined with business data, such as inventory, production, and maintenance scheduling, in order to create insights that lead to better business decisions. In this use case, the company given as example, is a leading manufacturer of innovative protective packaging materials and systems for medium-to-large manufacturers and distributors. Rather than selling machines, the company provides them under a consignment model [19]. The company is launching a new *smart model*

that provides data streams that allow for inventory replenishment, management, condition monitoring, and performance optimisation. Collecting and monitoring these data streams across all of the deployed machines will allow the company to:

- Decrease replenishment costs for consumables.
- Optimise service routing and scheduling based on actual usage patterns and analytics.
- Improve machine up-time and decrease machine downtime alerts through remote condition monitoring.
- Improve overall customer satisfaction.

For other industries, a similar solution might require a selection of modular components from the joint reference architecture, different to those described in this specific use case.

3.3. Edge Computing RA 2.0

In 2016 The Edge Computing Consortium (ECC) was created by several organisations: Huawei, the Shenyang Institute of Automation (SIA) of the Chinese Academy of Science, Intel, ARM, iSoft-Stone and China Academy of Information and Communications Technology (CAICT). On the other hand, the Chinese Academy of Telecommunications Research (CATR) together with the Ministry of Industry and Information Technology (MIIT) formed the Industrial Internet Alliance (IIA) in 2016 with the aim of boosting the development of industrial Internet in China [20].

The Edge Computing Reference Architecture (EC-RA) 2.0 was proposed from the joint work between the ECC and the AII and it is based on a horizontal layers model with open interfaces. However, vertically this architecture uses the following services: management, data life-cycle and security, with the aim of providing intelligent services throughout the life cycle. The development of this architecture was based on international standards such as ISO/IEC/IEEE 42010:2011 and it presents Edge solutions and frameworks to industries. Fig. 3 presents a horizontal perspective of the architecture with the following layers [20]:

- **Smart Services:** this layer is based on a model-driven service framework. Intelligent coordination between service development and deployment is achieved through the *Development service framework* and the *Deployment and operation service framework*. These frameworks enable coherent software interface development and automatic implementation and operations.

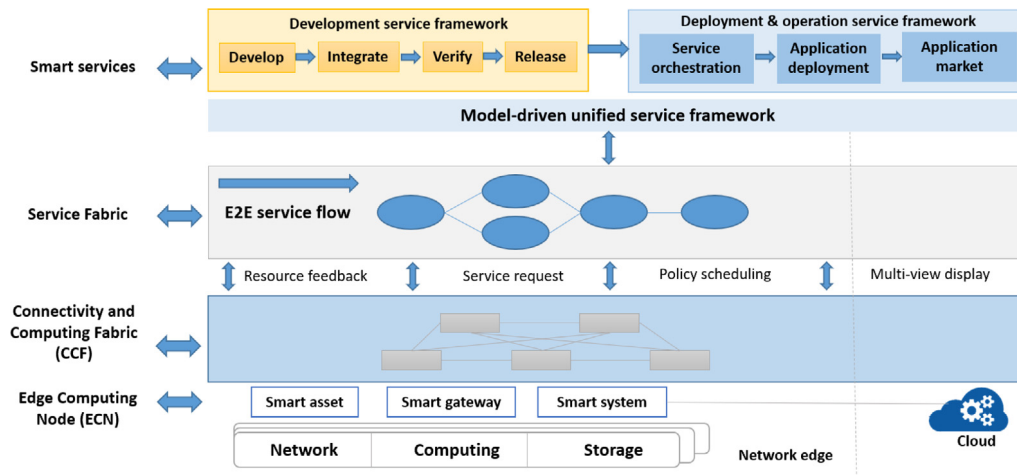


Fig. 3. Edge RA 2.0 (Edge Consortium). Based on work of Edge Computing Consortium, Alliance of Industrial Internet [20].

- **Service Fabric (SF):** defines the tasks, technological processes, path plans, and control parameters of the processing and assembly phases, implementing fast deployment of service policies and fast processing of multiple types of products.
- **Connectivity and Computing Fabric (CCF):** the Operation, Information and Communications Technology (OICT) infrastructure is responsible for deploying operations and coordinating between the computational resource services and the needs of the organisation.
- **Edge Computing Node (ECN):** in this layer the intelligent Edge Computing Nodes (ECNs) have real time processing and response capacities, are compatible with diverse heterogeneous connections and the security is integrated into the hardware and software.

The architecture was developed on the basis of the international standards defined by ISO/IEC/IEEE 42010:2011, which strengthens its use as a reference for the development of industrial solutions with Edge Computing.

3.3.1. Application scenario: Smart water

The Edge Computing Consortium proposed a Smart Water use case. In this case, a water supply system represents a key urban infrastructure since its correct functioning is essential for the construction of safe smart cities. Based on Edge Computing platforms, the smart water solution uses advanced sensing, networking, computing, control, and intelligence technologies to monitor equipment such as secondary water supply facilities. The urban water supply equipment, information system, and service processes are integrated to support the large-scale exchange of a vast amount of data between multiple systems. This enables full process control, automatic fault diagnosis and predictive maintenance, reducing energy consumption and ensuring water security. The core contributions of Edge Computing to the smart water solution [20] include:

- Smart gateways collaborate with Cloud-based platforms to receive data analysis models, so that they can analyse data and respond in real time to requests based on company policies.
- Edge devices are deployed in a distributed way and their number can be expanded on demand.
- The solution supports life-cycle management of multiple apps for the flexible addition of new services.
- Hardware platforms have an industry-grade design and multiple interfaces that comply with various protocols.

3.4. Industrial Internet Consortium RA

The Industrial Internet Consortium and the Edge Computing Consortium have developed their reference architecture using the ISO/IEC/IEEE 42010:2011. This standard is the principal guide to identifying conventions, principles and best practices for consistent Internet of Things architectures or frameworks. The ISO/IEC/IEEE 42010:2011 facilitates the development of evaluation, communication, documentation and systematic or effective resolution in a reference architecture [21]. Fig. 4 presents the Industrial Internet Consortium Reference Architecture (IIC-RA) and its three principal tiers: *Edge*, *Platform* and *Enterprise*. Those are described as follows [21]:

- The *Edge layer* collects data from the Edge nodes through a proximity network. The main architectural features of this layer include: breadth of distribution, location, scope of governance and nature of the proximity network. Each of the characteristics will change according to the use case.
- The *Platform layer*, is responsible for processing and sending control commands from the third layer (enterprise) to the Edge layer. Its function is to group the processes and analyse the data flows from the Edge layer and the upper layers. It manages the active devices in the Edge layer for data consultation and analysis through domain services.
- The *Enterprise layer* is located in the cloud and runs specific applications such as: decision support systems [50], end-user interfaces, and operations management. This layer generates control commands to be sent to the platform and Edge layers and also receives data flows from those layers.

3.4.1. Application scenario: Electric Submersible Pumps (ESP)

The Industrial Internet Consortium suggests as a use case a scenario aimed at predicting equipment failures and scheduling proactive response in Electric Submersible Pumps (ESPs). In this scenario, the machine learning models used to predict Electric Submersible Pump (ESP) failures, require data from multiple off-shore platforms. The analytic models are complex and a large amount of data is needed to train and re-train the models. They also require that regular data feeds from operating ESPs determine each unit remaining useful life. The data from individual ESPs need to be analysed regularly, but information decay is much slower than in the other scenarios and decisions that can be taken daily or weekly. Computation is typically performed at the enterprise level using a public or private Cloud and it is at the top end of the Edge continuum. The Edge can be anywhere

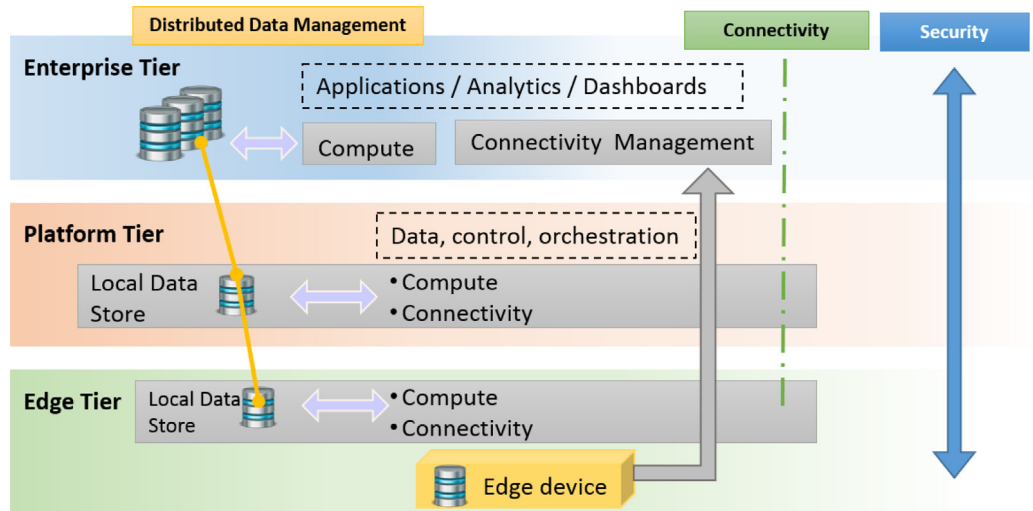


Fig. 4. Industrial Internet Consortium RA. Based on the work of Tseng et al. [21].

Table 3
Comparison of the main characteristics of the four RAs and of the one proposed in this work. The check-mark (✓) means advantages and x-mark (X) means weaknesses.

Characteristics	FAR-Edge	INTEL-SAP	ECC	IIC	Global Edge
Connectivity and communication	✓	✓	✓	✓	✓
Device Management	✓	✓	✓	✓	✓
Data collection, analysis and performance	✓	✓	✓	✓	✓
Scalability	✓	✓	✓	✓	✓
Standards	X	X	✓	✓	✓
Security	✓	✓	✓	✓	✓
Data Encryption	X	X	X	X	✓
Blockchain	✓	X	X	X	✓
Royalties or fees	X	X	X	X	✓
Open Source	✓	X	X	X	✓

along the time-value graph. It is *where* sensor data is used to achieve a specific key objective or to address a specific business problem [21].

3.5. Final remarks

The four reference architectures described in this section are based on a three-layer model for Edge Computing, which does not replace Cloud services, but integrates them into the final level of the architecture. In cases where the volume of data is high, the Edge nodes of these RAs are the first to process, control and reduce the volume of data transferred to services running in the Cloud. Thus, the Edge nodes reduce storage requirements, reduce latency and provide response in real time. Table 3 summarises the main characteristics of those reference architectures and compares them with those of the Global Edge Computing Architecture proposed in this work.

These RAs focus on Cloud and Edge Computing for industrial environments with the following expected outcomes: *increased use of data, decentralised control, improved automation and process security*. Although all of them offer numerous advantages to applications and environments in which large volumes of data are collected and processed, only FAR-Edge RA includes an intermediate layer called *Ledger*, based on the concept of *smart contracts*, which does not represent physical components and even the entities it contains are abstract. Services of this layer are transaction-oriented, where requests to modify or make changes

in a shared state of the system must be evaluated and approved by peer nodes before their execution. These ledger services consist of codes that are executed simultaneously on all active nodes at the time a service is requested [44].

4. Global Edge Computing Architecture: a reference architecture

This section presents the new *Global Edge Computing Architecture*, which includes a level of security that starts at the base layer by encrypting sensor-generated data which are further processed at the Edge layer. Fig. 5 illustrates the proposal. Blockchain technologies have been leveraged to strengthen the security provided by the proposed architecture. In 2008 Satoshi Nakamoto presented the blockchain concept as a mechanism that allows some actors to perform verified transactions [51]. The vision of Nakamoto was to provide a ledger with characteristics such as: *distribution, immutability, transparency, security and auditability* [51]. In addition, other authors such as Antonopoulos [23] or Reyna et al. [22] point out that blockchain allows to access transactions openly and fully through queries and all entities of the system which can be verified and collated at any time.

The design of the proposed architecture is based on some components of the previously analysed four main Reference Architectures that strengthen the RA and which are described as follows:

- *FAR-Edge*: The main component of this architecture is blockchain which is incorporated into the intermediate layer; the ledger services evaluate and approve of the service request calls made by the Edge nodes in the first layer. With that frame of reference, the proposed architecture incorporates blockchain into its first layer to protect data from the very moment it is generated by the sensors, as described in Section 4.1.
- *INTEL-SAP*: The SAP Cloud Trust Center is an important component because it verifies device ownership and registers it. It generates a certificate of authenticity and keeps the list of authorised devices up to date. The proposed architecture includes an authentication mechanism in the Cloud that validates the device or user access before they can participate in the systems and platforms built on the basis of the new RA.

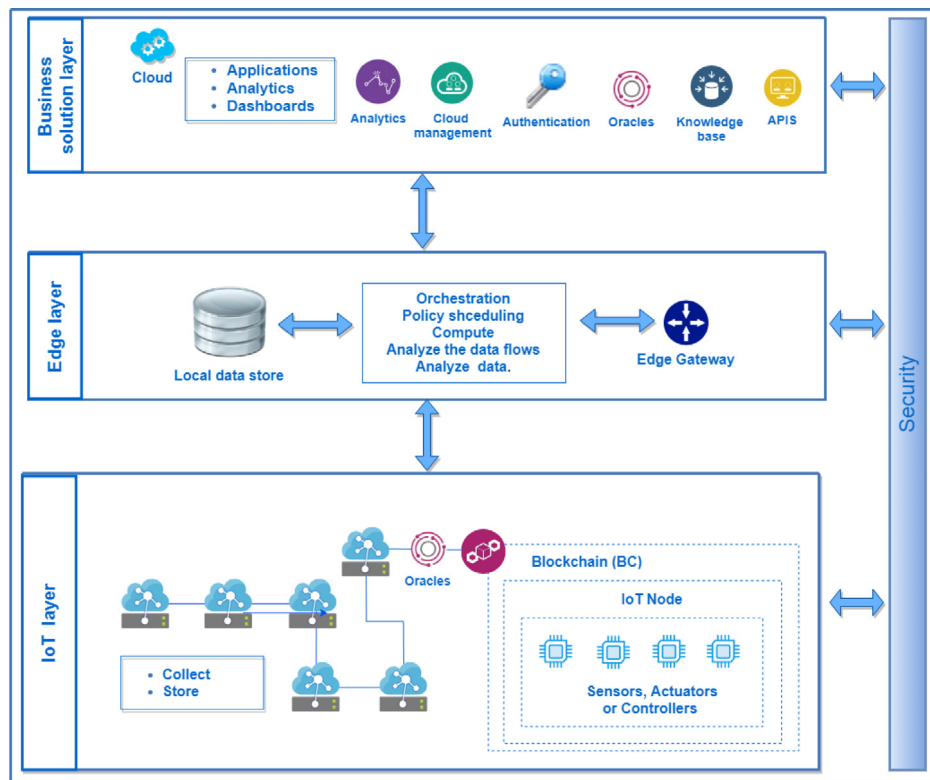


Fig. 5. Global Edge Computing Architecture.

- **Edge Computing RA 2.0:** The Edge Computing Consortium stresses the importance of using International Standards such as: ISO/IEC/IEEE 42020:2011 when presenting Edge solutions and frameworks to industries. Therefore, the proposal is based on the IEC 61499 standard, which proposes the design of architectures by means of functional blocks (FBs) defined as the basic structural unit of the models. Each block is characterised by its inputs, outputs and internal functions, as is the case of classic block diagrams. The FBs are not executed until they receive an input signal, so they remain in a state of *rest* in order to allow for portability and reuse of functional blocks, which behave as a completely independent unit from the rest [52]. Other authors have used blockchain technologies for Edge Computing, such as Stanciu [53], who proposed a blockchain-based distributed control system for Edge Computing, based on the IEC 61499 standard. He proposed the use of blockchain technologies in the last tier with Cloud services as a supervision strategy.
- **IIC: FAR-Edge RA** is based on this architecture. Authors of the present paper consider the enterprise layer an important element in which specific processes and applications in the Cloud (private or public) are executed, such as: decision support systems, operation management, big data analytics, model training, predictions and business analytics.

Fig. 5 depicts the proposed *Global Edge Computing Architecture*. It consists of three different layers which are described in depth in the following sections.

4.1. IoT layer

It is the first layer of the architecture and it is formed by the devices or physical objects that monitor services, activities or equipment in operation. Sensors, actuators, controllers and

gateways for IoT environments, are included in this layer allowing for the management and storage of computational resources in IoT devices. In IoT environments, a wireless standard is normally applied, such as Wi-Fi, BLE, ZigBee, LoRa or SigFox [54]. The principal concepts of those standards are as follows:

- **Wi-Fi:** intended for wireless communication in local areas. It uses spectral bands of 5 GHz and 2.4 GHz. Due to this limited range it is mostly used in closed spaces, providing Internet connectivity through nodes known as *wireless network access points*. Access points typically operate within a maximum range of 1 km. The main disadvantages of this technology are high energy consumption and the security risks associated with free WiFi connectivity. Another problem in the communication process is the frequency with which WiFi is used and its sensitivity to interference [55].
- **BLE:** Bluetooth Low Energy (BLE) is an enhanced version of the original Bluetooth Version 4.0, designed to manage large-scale devices with low latency and efficient power consumption, but only for short-range wireless applications. Similarly, Classic Bluetooth technology BLE operates in the ISM band from 2.4 GHz to 2.4835 GHz. [56].
- **ZigBee:** In 2005 the ZigBee Alliance introduced ZigBee as a technology based on the IEEE 802.15.4 standard for companies requiring low latency and low consumption communication. It is adaptable to other standard protocols and uses direct address assignment, so that each node knows where the packet should be sent. Currently, ZigBee is widely used in energy demand and load management scenarios [57] [15].
- **LoRa:** The LoRa Alliance introduced LoRa technology in 2015, it is based on a Chirp Spread Spectrum (CSS) physical layer in which each symbol is encoded with a longer bit sequence. In the LoRa network topology each final device communicates in a single hop with one or more of the LoRa gateways and these in turn with a central node called *LoRa NetServer*

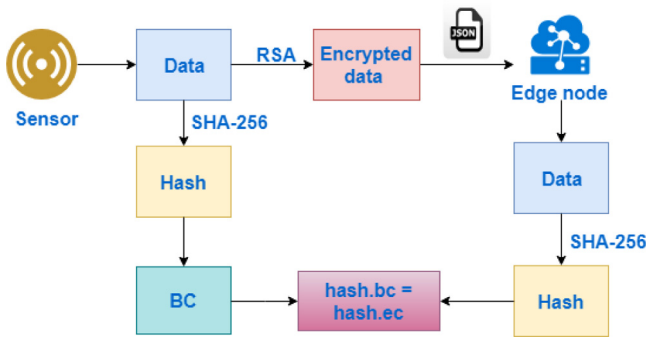


Fig. 6. Encrypted data.

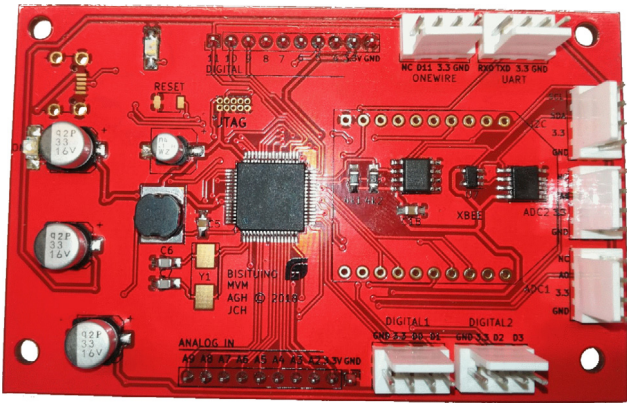


Fig. 7. Crypto-IoT board.

that functions as a server. When LoRa is used, an end device is never directly associated with the Gateway but with the NetServer, so all the complexity is transferred to the central node [58].

- SigFox: It is a network of low-power extended areas for Machine to Machine (M2M) applications such as security, healthcare and points of sale. SigFox's main objective is to use the UNB (Ultra Narrow Band) technology to connect remote devices that require the transfer of small amounts of data in low bandwidth and with efficient energy consumption. SigFox complies with M2M requirements for complex interface operations with low power requirements [55].

Blockchain is another key element of this layer. Samaniego and Deters [59] suggest that blockchain-based permissions are hosted in the Cloud to ensure their distribution and persistence in data storage. To guarantee the integrity and validity of the data in this layer, a basic blockchain scheme is used with hash values and the SHA-256 and RSA algorithm. SHA-224 and SHA-256 use the same sequence of sixty-four constant 32-bit words [60]:

$$K_0^{256}, K_1^{256}, \dots, K_{63}^{256} \quad (1)$$

In the case of SHA-256, the initial hash value, $h^{(0)}$ is formed by:

$$h^{(0)} = 6a09e667$$

$$h^{(1)} = bb67ae85$$

...

$$h^{(7)} = 5be0cd19$$

Hash functions are used when it is necessary to compress heterogeneous data (variable lengths or e-bits) and to which the hash value name h is assigned [61]. The SHA-256 can be applied to compress data in a message and obtain a message digest in a block [60,62]. Likewise, the RSA algorithm has been developed for public key encryption in 1970 by Ron Rivest, Adi Shamir and Len Adleman [63]. Stallings et al. [64] describe the algorithm and mention that it is based on a public key PU and a private key R . These keys consist of two prime numbers, $PU = e, n$ and $K = d, n$. To encrypt a message M a cipher C is calculated using $C = M^e \bmod n$. The message can later be retrieved using the private key $M = C^d \bmod n$.

Fig. 6 illustrates the process of protecting sensor data through the use of SHA-256 and RSA, ensuring their legitimacy, reliability and distributed identity with authentication and authorisation mechanisms [22]. In the proposed architecture, the IoT node or device is assigned a public key. The SHA-256 is applied to the generated data to create a hash that is stored in the blockchain. Moreover, the RSA algorithm encrypts the data and sends it to an Edge node as a JSON object. The Edge node is going to use a

private key to decrypt them. At this point in the procedure the Edge node can apply a SHA-256 that is going to create a hash. If there are doubts or indications of attacks or malicious data, the hash generated by the Edge node can be compared to that stored in the blockchain. If the hashes are equal the data is correct.

The concept of *oracles* is included in this layer. Oracles are agents that work as brokers between the blockchain and the data coming from the real world. Oracles verify data and send them to the blockchain, where they can be used in Smart Contracts. The role of the oracles is to provide external data securely and in compliance with the predefined conditions established in the Smart Contracts, unlocking the values and executing the events in the blockchain. The values or data sets can be of any type, such as: temperature, price fluctuations, humidity, vibrations, payments records, etc. [65].

The IoT oracles (that is, I/O sensor or actuator ports) and the blockchain interact with each other in the Global Edge Computing Architecture thanks to the Crypto-IoT boards shown in Fig. 7, they have been designed and mounted by the BISITE Research Group of the University of Salamanca. These devices are designed as generic gateways intended for building applications and systems based on the *blockchained* IoT paradigm. The strong point of these data-agnostic devices is that they ensure the security and reliability of the data they obtain from the environment and transfer it to the upper layers of the developed architecture, with which they are totally agnostic.

The features of the Crypto-IoT devices are important for the overall functioning of the Global Edge Computing Architecture, these include:

- ATSHA204A encryption chip with SHA-256 technology.
- XBee for the integration of different telecommunication modules (ZigBee, Wi-Fi, Bluetooth, LoRa, FM Radio, RPMA).
- A USB communication module acting in host mode which allows for the connection of micro-servers such as Raspberry Pi and neural processing modules.
- All kinds of I2C sensors in a *plug-and-play* way.

4.2. Edge layer

This layer is located near the Edge endpoint nodes allowing for the correct orchestration of the different technological assets in the organisation, contributing to a significant improvement in the supply, monitoring and updating of existing resources. The technological support on which this layer is based includes the management of critical business activities. These activities include the management of physical resources (sensors, actuators

and controllers) or the study and analysis of large volumes of data in real time.

This layer is central to the Global Edge Computing Architecture, because it is responsible for the filtering and pre-processing of the data that is obtained from the IoT layer and is transferred to the layers in the Cloud (known as the *Business solution layer* in the proposed Global Edge Computing Architecture). It maintains the integrity of all the data obtained from the IoT layer, thanks to the blockchain implemented in the lowest layer of the architecture. The data collected by the sensors deployed in the IoT nodes, usually ruled by micro-controllers with reduced computational and storage resources, are filtered and pre-processed in computing elements within the Edge layer.

These elements can be low-cost solutions such as micro computers based on educational architectures like Raspberry Pi [66] or open architectures like Orange Pi which are based on Raspberry Pi. These devices have a cost of a few tens of US dollars and provide Linux architectures of 32 to 64 bit, with between 256 MB and 1 GB of RAM, microprocessors of between 1 to 4 cores and clock frequencies at around 1 GHz, allowing for the use of SD cards with several GB of storage. That is, they allow to process a much greater amount of information than the micro-controllers that are habitually used in IoT devices, aimed at minimising energy consumption and executing simple reading logic and control programs on general purpose input and output ports (GPIO), serial ports (UART, SPI) or communication buses between integrated circuits (I2C).

Although this kind of devices also come with I/O ports that allow to connect them with sensors and actuators, making possible the implementation of the Edge and IoT layers within a single device. The general approach is to use them to coordinate a network of cheaper IoT devices, where they act as hub elements or data collectors throughout the IoT and up to the Edge. These devices are sometimes located beyond the reach of power supply networks and must therefore be powered by batteries and solar panels.

The proposed architecture is open to Edge devices running new IoT layers from different providers, including Amazon Web Services IoT Greengrass, Microsoft's Azure IoT Edge or Google's Cloud IoT Edge [67]. These layers make it possible to apply Machine Learning techniques in the Edge, for instance by means of TensorFlowLite libraries [66] in Raspberry Pi or similar devices, thus facilitating the performance of Data Analytics in the Edge, saving the amount of data sent to the Cloud, reducing its associated costs, as well as offering valuable data to users even if communication with the Internet and the Cloud are temporarily lost.

4.3. Business solution layer

This layer includes the set of services and business applications present in the Cloud. Here individual API calls are activated by executing more complex sets of operations that involve the use of interactive interfaces and are part of the business applications ecosystem. Public (hosting data on commercial servers) or private (corporate data centre) Cloud services can be used at this level. The main components of this layer include:

- *Analytics*: the use of case based reasoning, artificial intelligence techniques and machine learning algorithms allows for greater flexibility in the data analysis and visualisation capabilities required by different business units and operating teams.
- *Cloud management*: a storage and administration service allows for physical or virtual segregation of the stored data according to the tenant being able to track the use of the

service by the tenant. The use of the service by several tenants is an integral feature even in private cloud management services. In this context the tenants will be the different departments or working groups of a public or private organisation. Scalability is another distinguishing feature of the Global Edge Computing Architecture and from a storage standpoint, it is often derived from technologies that leverage large groups of consumer hardware components that can be easily and economically expanded as the demand for infrastructure and storage resources grows.

- *Authentication*: there are two ways in which the Global Edge Computing Architecture addresses authentication, authorisation and distributed transaction:
 - In a typical Smart City context, the Cloud-Edge is public and many different entities may participate in it. The central authorisation entity (Authentication) in the cloud establishes a smart contract with each new IoT element to allow it to transfer or extract information from the blockchain and determine the conditions under which the process is going to be carried out.
 - When the Cloud-Edge is a private environment it uses a *permissioned blockchain*. The initial node that has created the blockchain (the central authorisation entity in the cloud) establishes the rules in the first block, in this way, only the administrator nodes can assign new nodes the role of an administrator. Only these nodes can provide access to new IoT nodes, moreover they are in charge of transferring or extracting traffic from the blockchain. By means of a consensus algorithm, the acceptance process is executed between the administrator nodes and the devices. This implementation applies to case studies such as Industry 4.0 [68] as demonstrated in the agroindustry case study described in Section 5.
- *Knowledge base*: a social machine could be developed using virtual organisation of agents and support decision systems based on sensors data [69]. This component will be complemented by cloud-based orchestration that will enable the provisioning, monitoring and updating of connected technological resources.
- *APIs*: it is a set of applications (with standard methods, such as HTTP RESTful, XML and SOAP calls [70]) through which cloud services can be called. Thanks to these applications, services can be made available through a standard web browser or other HTTP client application.

5. Use case: Implementation in a real scenario of an agroindustry platform based on the new architecture

The new industry oriented Edge Computing architecture has been tested with the developed agroindustry platform designed to monitor and optimise the management of agricultural and livestock farms. The main objectives of the use case is to maximise the platform's benefits, minimise its expenses and save time and money through efficient management of a mixed dairy farm in Castile and León, Spain. Once the platform has been implemented, it was possible to locate and identify the available resources in this agroindustry scenario. As a result, the resources have been used more efficiently. It has been possible to locate the animals and monitor their health conditions in real time. The application of Data Analytics and Machine Learning techniques has facilitated the diagnosis because it allowed to associate the obtained parameters with specific diseases.

The proposed Edge architecture has been designed to manage at the Edge the information received from the installed sensor network. It provides the platform with the following features:

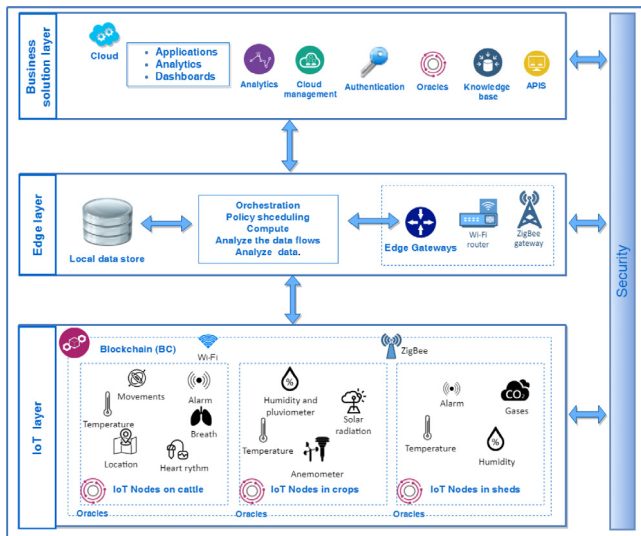


Fig. 8. The Agroindustry platform based on the new Global Edge Computing Architecture.

- The new architecture has a high availability environment that provides users with real-time information by means of the Business Solution layer, which the users can access remotely.
- New sensors can be incorporated into the open sensor platform through the architecture's IoT and Edge layers.
- Both the Edge and the Business Solution layers include Artificial Intelligence techniques such as Case Based Reasoning algorithms for the detection of anomalous patterns in the behaviour of the livestock.

5.1. Requirements and layer specifications

Each layer has a different role designed to meet the initial mixed farming requirements (see Fig. 8):

- **IoT Layer:** consists of a set of heterogeneous IoT networks [70] which are used for sensitisation and performance monitoring. This layer facilitates the interaction between all the systems and the environment. It includes all the sensors needed for the collection of contextual information related to dairy cows, barn conditions and the crops used to feed the livestock.
- **Edge Layer:** a management, planning and monitoring platform that collects and pre-processes the information at the Edge for further processing in the Cloud. Special collecting nodes with crypto IoT chips are used to establish a distributed ledger and implement blockchain. Moreover, this layer performs the first *pre* Data Analytics filtering to reduce transmission costs when data are sent to the upper Cloud layers.
- **Business Solution Layer:** The architecture's Business Solution Layer contains a Cloud Computing platform whose virtual agent organisation develops a social machine [71]. This social machine provides support in decision making on the basis of the parameters collected by the sensor network. The platform provides data visualisation methods and an alert management system according to the monitoring of all the heterogeneous IoT sensor networks.

The platform is based on the new Global Edge Computing Architecture, the proposed approach is versatile and simple to

manage. A set of sensors deployed on the farm measure parameters related to the livestock and the crops. The architecture visualises in real time the values collected by those sensors, contributing in this way to effective decision making. We underscore that the developed platform is open and inter-operable, key features that make it possible to add other technologies to it at a later stage for the purpose of improving its performance, accuracy and data flow.

5.2. IoT layer

There are three heterogeneous IoT sensor networks in this scenario which monitor different elements of the environment:

- **Crops:** the platform includes several sensors that monitor the ambient conditions of the crops used as feed for the livestock (independently of the type of crop: corn, alfalfa and rye). Their measurements allow to take optimal irrigation decisions:
 - Thermometer (measures air temperature).
 - Hygrometer (measures the relative humidity of the air).
 - Pluviometer (measures rainfall and artificial irrigation).
 - Anemometer (measures wind speed and direction).
 - Soil thermometer (measures ground temperature).
 - Soil moisture sensor (measures ground humidity).
 - Solar radiation sensor (used to estimate evapotranspiration levels).
- **Barn sensors:** This group includes all the sensors that are used in the buildings where dairy cows live:
 - Thermometer (measures air temperature).
 - Hygrometer (measures the relative humidity of the air and estimates temperature–humidity index, related to dairy cows stress [72]).
 - Gas sensors (detect hazardous levels of gas in the air): Hydrogen sulphide, Carbon dioxide, Ammonia and Formaldehyde.
- **Dairy cow sensor:** This set includes all the bio-metric sensors responsible for taking measurements of parameters directly related to the cows' bodies, detecting possible fevers, stress and other problems or illnesses.
 - Body temperature.
 - Heart rate.
 - Breathing rate.
 - Gyroscopes that measure the inclination of the animal's head, allow to find out if the cow is resting, eating or ruminating.

To connect the wireless devices listed in this scenario (sensors and IoT gateways), the ZigBee communication standard [54] has been used. IoT gateways gather information sent by the sensor nodes and forward it to the micro computers that form the Edge layer. These micro computers filter, pre-process and transfer this information through the Internet (using cellular networks or DSL) to the Business Solution layer.

Fig. 9 shows the plan of the area in Castile and León (Spain), chosen for the implementation of the system. This Figure shows the location of the barns and the offices, where the following nodes have been deployed:

- **Node 0:** IoT gateway placed in the office building. It collects data coming from the sensor nodes in both barns as well as the closest agrometeorological stations. This gateway transfers data to the Cloud using an Ethernet/DSL Internet connection.



Fig. 9. Deployment of the IoT sensor and Edge nodes in the barns and the IoT agrometeorological stations in the associated crops.

- Node 1: sensor node in barn #1.
- Node 2: sensor node in barn #2.

In the case study crops there are also two complete *agrometeorological base stations* (nodes 100 and 200). The rest of the nodes are *agrometeorological satellite stations* (i.e., non-base stations with fewer sensors) that measure only precipitation–irrigation and soil moisture and are placed at shorter distances from each other than the base stations.

Fig. 10 shows a functional design scheme that illustrates the location of the wireless sensor nodes in every barn (top) as well as their physical appearance in the laboratory before they had been installed in the real scenario (bottom). In addition to the power supply module, the following analog sensors are incorporated, which are digitised thanks to an analog–digital converter module with four input channels and passing digital information to the micro-controller through I2C hardware protocol [73].

- MQ-4: CH₄: methane
- MQ-136: H₂S: hydrogen sulphide
- MQ-137: NH₃: ammonia
- MQ-138: CH₂O/H-CHO/formaldehyde

Moreover, a SHT20 temperature probe collects and digitises temperature and humidity values, by combining these values we obtain the temperature–humidity index; high values are indicative of stress in dairy cows [72].

There are two types of agrometeorological IoT stations for crop monitoring: complete base stations and satellite stations. Complete base stations are capable of measuring all the parameters on the field because they incorporate all the required sensors. Satellite stations (i.e., non-base stations) have reduced capabilities and only incorporate the sensors that are used to measure precipitation/irrigation and soil moisture. This is because these physical quantities vary over short distances. The measurements collected by satellite stations are sent to the base stations using ZigBee transmission, which in turn forward them to the Edge layer at the main offices. Thus, the complete base stations include:

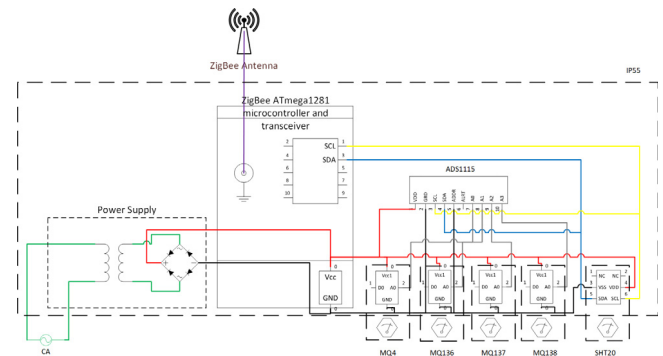


Fig. 10. Functional blocks of the sensor nodes (top) and physical aspect in laboratory before installation in barns (bottom).

- Solar panel.
- Battery for energy storage.
- Power supplied by the panel and battery.
- Battery level sensor.
- The previously described crop sensors.
 - Thermometer (not present at satellite stations).
 - Hygrometer (not present at satellite stations).
 - Pluviometer (measures both rainfall and artificial irrigation).
 - Anemometer (not present at satellite stations).
 - Soil thermometer (measures ground temperature).
 - Soil moisture (measures ground humidity).
 - Solar radiation (not present at satellite stations).
- General purpose micro-controller.
- IoT Crypto chip for blockchain features.
- ZigBee module for data collection from satellite stations and forward to Edge Layer in main offices.

Fig. 11 shows the functional block diagram of a complete agrometeorological base station (left-hand side), as well as the physical aspect of a satellite station in the real alfalfa crop monitoring scenario (right-hand side). Finally, there are IoT devices for the monitoring of bio-metric indicators in livestock. The sensors for measuring parameters related to dairy cows include the following elements:

- 5 V Battery charge.
- Microprocessor for obtaining, processing and sending information.
- Wi-Fi for the transmission of information to the Edge.
- Sensors: temperature, respiration, heart rate and accelerometers.

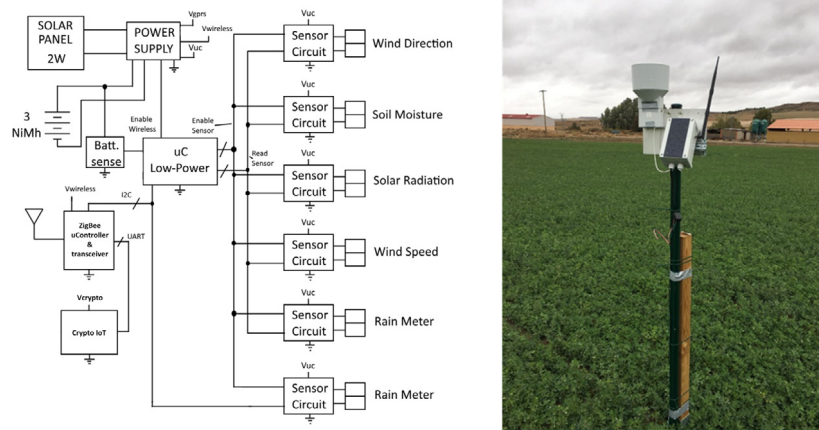


Fig. 11. Functional design diagram of the complete agrometeorological base station (left-hand side) and the physical aspect of an agrometeorological satellite station (non-base) in a real alfalfa crop monitoring scenario (right-hand side).

5.3. Edge layer

Fig. 12 depicts the functional design of the IoT gateway in the Edge layer which has been implemented in the office building located between the two barns. The figure shows the layer's power supply, the micro-controller module with the ZigBee transceiver, the IoT crypto chip (belonging to the IoT layer in the Global Edge Computing Architecture), blockchain features and the Ethernet connection expansion module for data tunnelling between the blockchain data and the Edge node which filters and pre-processes data before forwarding it to the Business Solution layer via Internet.

The ZigBee gateway at the Edge is capable of incorporating data collection technologies other than ZigBee, such as LoRa. Furthermore, it can tunnel information through Wi-Fi or GPRS instead of Ethernet.

The Edge node that filters and pre-processes the IoT data before forwarding it to the Business Solution layer in the Cloud, is formed in this use case by a Raspberry Pi 3 Model B with the following computing, storage and communication characteristics:

- CPU: Broadcom BCM2837 SoC (ARMv8-A 64/32-bit instruction set), with 4 cores Cortex-A53 at 1.2 GHz.
- RAM: 1 GB (shared with GPU).
- Storage: 16 GB microSD card (10 MB/s).
- I/O: 4 USB (via on-board 5-port USB hub), 17 GPIO plus some specific functions such as I2C, UART and SPI.
- Networking: 10/100 Mb/s Ethernet, 802.11b/g/n single band 2.4 GHz wireless and Bluetooth 4.1 BLE.

The Raspberry Pi is used as an Edge node, it filters and pre-processes data by means of TensorFlowLite libraries [66] over a Node.js based server that gathers data coming from IoT sensors. The node eliminates noise and serves summarised information through a Web interface and forwards value-added data to the Business solution layer in the Cloud.

5.4. Business solution layer

Finally, one of the most important components of the use case is the online platform that runs on the remote virtual server in the Business solution layer. This online platform is accessed by users. Fig. 13 shows a screenshot of this application accessed from a web browser and the monitored bio-metric values in a sample dairy cow. The interface in the figure displays two graphs that provide location and health information of a specific cattle

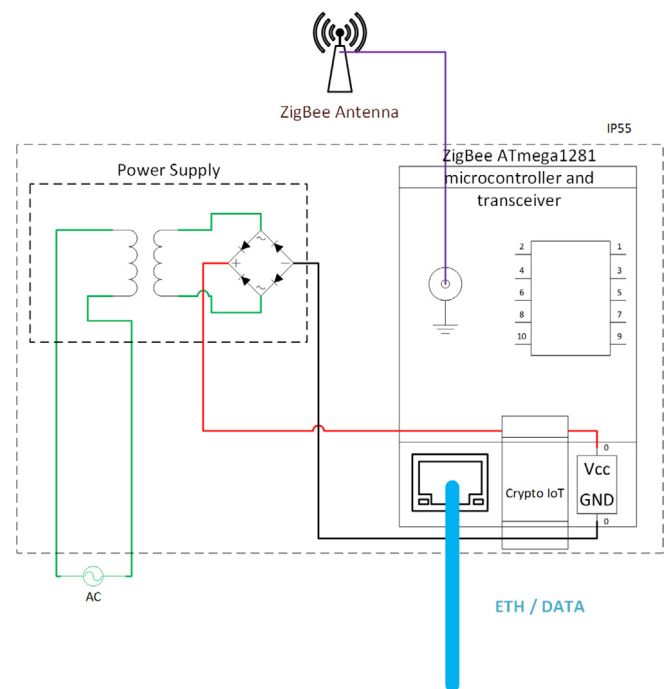


Fig. 12. Functional design of the IoT data collection gateway at the Edge layer.

selected by the user. The graph in the upper part of the interface shows an index that represents the activity level of the cattle over a period of time, it is calculated on the basis of its movements which are recorded by the accelerometers deployed on its collar. The graph in the lower part of the interface indicates the cattle's temperature, heart rate and breathing rhythm over time, also coming from the sensors attached to the cattle's collar.

5.5. Experiments and results

To evaluate the Global Edge Computing Architecture which has been implemented in the use case platform, two one-month tests were carried out in separate time periods. The first test was used as a baseline period while the second one was used to verify the possibility of optimising the performance of the same system through the addition of the Edge layer. The Edge layer aims to facilitate the transfer of a larger amount of data between the

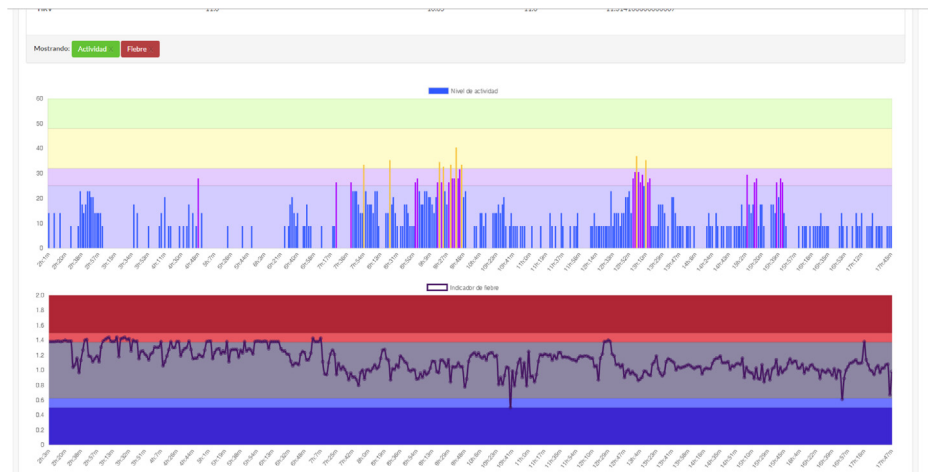


Fig. 13. Screenshot of the dairy cow monitoring application over the agroindustry platform running on the Business Solution layer.

sensors and the Cloud. The two test were subjected to exactly the same conditions. The importance of verifying the feasibility of the Edge layer lies in the fact that data transfer represents a large part of the total cost of system and application use in the Cloud. In the first test we measured the amount of data transferred in the old version of the system between the IoT sensors and the Cloud. This version did not benefit from the advantages provided by the Edge layer of the new reference architecture. In the second test, the same experiment was performed to evaluate the new version of the monitoring system which included the new Edge layer. Therefore, in the second test data transmission was measured in two stages: from the IoT sensors to the Edge layer, and from the Edge layer to the Cloud.

The system deployed in the first test was a prior version, it did not include the new Crypto-IoT devices, Edge elements based on the Raspberry Pi 3 Model B and TensorFlow Lite or blockchain. Thus, in the first test, the collecting nodes gathered the data through ZigBee and Wi-Fi and forwarded them directly to the application in the Cloud without filtering, pre-processing, or encrypting them.

In the second test, both the Crypto-IoT devices and the new Edge elements based on Raspberry Pi 3 Model B and TensorFlow Lite were incorporated.

In both phases, the same number of IoT nodes have been used:

- 1 Wi-Fi router gathering data from
 - 15 collars for livestock sensitisation (temperature, pulse, respiration and accelerometers).
 - * Every 5 min, the collar sends four values, one for each of the parameters it measures.
- 1 ZigBee collector gathering data from
 - 2 IoT nodes (temperature, humidity and four types of gases) in the two barns.
 - * Every node sends each of its values every 10 min.
 - 12 agrometeorological stations:
 - * 2 base stations: rain gauge, soil moisture, soil temperature, air temperature, air humidity, solar radiation, direction and wind speed.
 - * 10 satellite stations: rain gauge, soil moisture, soil temperature.
 - * Every station sends each of its values every 15 min.

Table 4

Data transmitted between the different Edge and Cloud components in the two tests.

Stage	Stage 1 (non Edge)	Stage 2 (Edge based)
IoT devices ↔ Edge nodes (free)	0.0 MB/month	98.2 MB/month
IoT devices ↔ Cloud	124.6 MB/month	0.0 MB/month
Edge nodes ↔ Cloud	0.0 MB/month	76.2 MB/month

Specifically, we measured the amount of data sent to the application layer in the Cloud, this layer was deployed in the Google Cloud platform using the App Engine as PaaS (Platform as a Service), Google Functions as Faas (Functions as a Service), Cloud SQL as MySQL relational database and BigQuery as the NoSQL database for the analysis of data in the Cloud, obtaining the following transmission measurements for the two tests, as represented in Table 4.

The data in the table demonstrates that the new reference architecture allows to build platforms in which the amount of total data transferred to the Cloud (in a system with the same conditions of use and sensitisation) was reduced by 38.84%. This reduction could be greater in other applications and systems that allow for a more beneficial use of Edge filtering and pre-processing.

Logically, in a massive data scenario the reduction in traffic also implies a reduction in costs associated with the transmission of data to the Cloud, the execution time of the computing components, as well as storage, mostly if it is over the free tier limit in the Cloud provider, due to the fact that redundant data have already been discarded on the Edge (e.g., retransmissions in the application layer of the ZigBee protocol stack).

Moreover, at a qualitative level, the implementation of the new architecture implied a substantial improvement in the security and integrity of the data. The old version of the system collected and encrypted data by means of a simple flow-based encryption algorithm based on a symmetric key of only 32 bit executed by ATmega 1281 8-bit microcontrollers and 8 kB of RAM. The encryption of the new schema is more secure thanks to the transmission of data through blockchain and encryption algorithms of SHA-256 which are implemented in Crypto-IoT devices. Furthermore, thanks to the possibility of accessing the Web server hosted on the Raspberry Pi, the farmer can visualise information, measurements and statistics from the last 30 days even if the connection with the Cloud has temporarily been lost due to limited access to Internet (information is transmitted via radio because of the lack of DSL or cable access).

In the first scenario, the IoT sensors send the data directly to the Cloud, which must bare the entire computational load of the application. In this case, the Cloud is where erroneous data are managed, duplicated data are discarded, and where operations such as averaging, identification of maximum or minimum peaks, application of other filtering algorithms etc., are performed. This implies that this kind of deployment requires greater computational, memory and storage resources in the Cloud, which imply higher monthly recurring costs, in comparison with a mixed Edge and Cloud deployment.

The use of an Edge-based distributed computing paradigm in IoT and Industry 4.0 scenarios, allows to move the computational load to the nodes in layers that are located closer to the data collection points. Thanks to the availability of low-cost and low energy consumption micro-computers in the market (such as Raspberry Pi or similar), it is possible to move part of the computational load to industrial facilities where data is being collected, reducing the traffic of the data sent to the cloud and saving the monthly recurring costs. In the new scenario, the platform continues to store the data in the Cloud which ensures its security and reliability. The difference is that the new Crypto-IoT modules deployed in the Edge preprocess and cipher the data before sending it to the Cloud. This does not put the data at risk, on the contrary, it provides the platform with extra security.

In this regard, security is one of the strongest features of the new reference architecture, since it provides a perfect protocol for securing data transfer between all entities, regardless of whether they are in the Edge or in the Cloud. Thanks to the implementation of the IoT layer within the proposed reference architecture, all communications are encrypted. Furthermore, its innovation in communications lies in the fact that no node in the network can repudiate the information provided by another node in it. Given that all the transmissions become part of the blockchain, the information remains inviolable allowing for the establishment of smart contracts among the different participating entities, including suppliers or customers within the industry value chain.

6. Conclusions

The current growth of industries is the result of the irruption of new technologies such as sensors, intelligent devices and objects, tablets, telephones, computers and others. This in turn has contributed to a surge in the amount and variety of data, leading companies to adopt technological solutions whose architectures are more robust, facilitating access to data and visualising data from multiple connected (wireless or wired) systems. Industries aim to reduce costs, achieve faster access and distribution times, improve or implement new services and technologies that meet the standards of quality, reliability, supervision and security demanded by an increasingly dynamic and competitive market.

This work has reviewed the most prominent reference architectures, proposed by influential technology manufacturers. All of the presented RAs are based on the ISO/IEC/IEEE 2010:2011 and their focus is the use of Edge Computing as a method of strengthening the capabilities of their Cloud-based technology implementations. This area of research has been significantly driven by the development and popularisation of trends such as IoT and machine to machine communication (Machine-Type Communications or MTC) which promote the generation of data, greater geographical independence in storage, immediacy in access and new forms of distribution and presentation.

On the basis of the existing reference architectures, a new RA has been developed with a modular and tiered design. The proposed Global Edge Computing Architecture allows to manage

complex solutions aimed at a wide range of IoT environments such as Smart Cities, Industry 4.0, smart energy, smart farming or healthcare, among many others. The new architecture has been evaluated in this respect through its application in an agroindustry use case conducted in the Community of Castile and León, Spain. Its results have demonstrated the advantages of the developed architecture in the following aspects: real-time analysis of data at the level of local devices and edge nodes and not in the cloud; lower operating and management costs due to reduced traffic and data transfer to the cloud; better application performance, as applications that tolerate latency can reach lower levels of latency at the edge of the network than in the Cloud; higher level of security through the incorporation of blockchain technologies into the IoT and business solution layers of the architecture.

Data Analytics algorithms based on Machine Learning models become key elements of applications for Industry 4.0 or smart farming scenarios where they extract valuable information from the data obtained from IoT devices. Usually, the learning and execution of these models implies a high computational cost, requiring calculation dedicated servers or powerful graphic cards. For this reason, one of the greatest challenges faced by these algorithms is the volume of information they must support. One of the ways of dealing with this problem is the implementation of Edge Computing technologies which perform information processing closer to the devices that generate information. These small Edge/IoT devices usually have limited capacities and limited battery life, so it is not reasonable to use their hardware intensely. It may be difficult to distribute processing efficiently in an Edge Computing system. In this sense, the proposed reference architecture provides added value because it considers the automatic balancing of computational load.

In a future work we are going to develop a new platform based on the Global Edge Computing Architecture. We are going to implement it in different scenarios in which the use of IoT data is massive, including smart cities, intelligent energy management and healthcare. Moreover, other experiments will be performed to evaluate not only the amount of data that is transferred to the Cloud, but also the computation time and the execution cost of the applications in the Cloud in the different situations: before incorporating the Edge layer and after implementing it through platforms based on the proposed reference architecture. Furthermore, we are going to test in depth the benefits of the proposed architecture and we are going to perform a comparative study of the new RA and other existing RAs.

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Conflict of interest

The authors declare that there is no conflict of interest regarding the publication of this manuscript.

References

- [1] N. Hassan, S. Gillani, E. Ahmed, I. Yaqoob, M. Imran, The role of edge computing in internet of things, *IEEE Commun. Mag.* 99 (2018) 1–6.
- [2] C. Stergiou, K.E. Psannis, B.-G. Kim, B. Gupta, Secure integration of iot and cloud computing, *Future Gener. Comput. Syst.* 78 (2018) 964–975.

- [3] F. De La Prieta, J.M. Corchado, Cloud computing and multiagent systems, a promising relationship, Springer, Cham., 2016, pp. 143–161.
- [4] T. Taleb, K. Samdanis, B. Mada, H. Flinck, S. Dutta, D. Sabella, On multi-access edge computing: A survey of the emerging 5g network edge cloud architecture and orchestration, *IEEE Commun. Surv. Tutor.* 19 (3) (2017).
- [5] R.N. Calheiros, R. Ranjan, A. Beloglazov, C.A. De Rose, R. Buyya, Cloudsim: A toolkit for modeling and simulation of cloud computing environments and evaluation of resource provisioning algorithms, *Softw. - Pract. Exp.* 41 (1) (2011.) 23–50.
- [6] M. Villari, M. Fazio, S. Dustdar, O. Rana, R. Ranjan, Osmotic computing: A new paradigm for edge/cloud integration, *IEEE Cloud Comput.* 3 (6) (2016) 76–83.
- [7] Nokia Solutions and Networks, Increasing Mobile Operators Value Proposition With Edge Computing, Technical Report, NOKIA Solution and Net, 2013. Available www.intel.com/go/commsinfrastructurehttp://nsn.com/portfolio/liquid-net/intelligent-broadband-management/liquid-applications (Accessed 20 november 2018).
- [8] M. Satyanarayanan, The emergence of edge computing, *Computer* 50 (1) (2017) 30–39.
- [9] Cisco, Cisco Global Cloud Index: Forecast and Methodology, 2016–2021, 2018. Available online: [http://www.cisco.com/en/US/solutions/collateral/ns341/ns525/ns537/ns705/ns1175/Cloud_Index_White_Paper.html\(#wp9000816\)](http://www.cisco.com/en/US/solutions/collateral/ns341/ns525/ns537/ns705/ns1175/Cloud_Index_White_Paper.html(#wp9000816)).
- [10] W. Shi, J. Cao, Q. Zhang, Y. Li, L. Xu, Edge computing: Vision and challenges, *IEEE Internet Things J.* 3 (5) (2016) 637–646.
- [11] D. Evans, The internet of things: How the next evolution of the internet is changing everything, 2011. Available online: http://www.cisco.com/web/about/ac79/docs/innov/IoT_IBSG_0411FINAL.pdf (Accessed 20 november 2018).
- [12] P. Garcia, A. Montresor, D. Epema, A. Datta, T. Higashino, A. Iamnitchi, M. Barcellos, P. Felber, E. Riviere, Edge-centric computing: Vision and challenges, *ACM SIGCOMM Comput. Commun. Rev.* 45 (5) (2015.) 37–42.
- [13] B.L. Risteska Stojkoska, K.V. Trivodaliev, A review of internet of things for smart home: Challenges and solutions, *J. Cleaner Prod.* 140 (2017) 1454–1464.
- [14] P. Chamoso, A. González-Briones, S. Rodríguez, J.M. Corchado, Tendencies of technologies and platforms in smart cities: A state-of-the-art review, *Wireless Communications and Mobile Computing* 2018 (2018).
- [15] O. García, R.S. Alonso, J. Prieto, J.M. Corchado, Energy efficiency in public buildings through context-aware social computing, *Sensors* 17 (4) (2017) 826.
- [16] K. Ha, Z. Chen, W. Hu, W. Richter, P. Pillai, M. Satyanarayanan, Towards wearable cognitive assistance, in: *Proceedings of the 12th Annual International Conference on Mobile Systems, Applications, and Services - MobiSys '14*, 2014., pp. 68–81.
- [17] B.-G. Chun, S. Ihm, P. Maniatis, M. Naik, A. Patti, Clonecloud: Elastic execution between mobile device and cloud, in: *Proceedings of the Sixth Conference on Computer Systems*. ACM, (11) 2011., pp. 301–314.
- [18] P. FAR-EDGE, FAR-EDGE Project H2020, 2017. Available online: <http://far-edge.eu/#/>.
- [19] INTEL-SAP, IoT Joint Reference Architecture from Intel and SAP, Technical Report, INTEL-SAP, 2018. Available online: <https://www.intel.com/content/dam/www/public/us/en/documents/reference-architectures/sap-iot-reference-architecture.pdf>.
- [20] Edge Computing Consortium, Alliance of Industrial Internet, Edge Computing Reference Architecture 2.0, Technical Report, Edge Computing Consortium, 2017. Available online: <http://en.econconsortium.net/Uploads/file/20180328/1522232376480704.pdf> (Accessed 20 november 2018).
- [21] M. Tseng, T.E. Canaran, L. Canaran, Introduction to Edge Computing in IIoT, Technical Report, Industrial Internet Consortium, 2018. Available online: https://www.iiconsortium.org/pdf/Introduction_to_Edge_Computing_in_IIoT_2018--06--18.pdf (Accessed 20 november 2018).
- [22] A. Reyna, C. Martín, J. Chen, E. Soler, M. Díaz, On blockchain and its integration with iot. challenges and opportunities, *Future Gener. Comput. Syst.* 88 (2018) 173–190.
- [23] A.M. Antonopoulos, *Mastering Bitcoin : programming the open blockchain*, second ed., " O'Reilly Media, Inc.", 2017., p. 371.
- [24] ISO/IEC/IEEE 42010, Systems and software engineering - engineering, Technical Report, ISO/IEC/IEEE 42010, 2011.
- [25] H. Gupta, A. Vahid Dastjerdi, S.K. Ghosh, R. Buyya, Ifogsim: A toolkit for modeling and simulation of resource management techniques in the internet of things, edge and fog computing environments, in: *Software - Practice and Experience*, Vol. 47, 2017., pp. 1275–1296.
- [26] X. Zeng, S.K. Garg, P. Strazdins, P. Jayaraman, D. Georgakopoulos, R. Ranjan, IOTSim: a cloud based simulator for analysing iot applications, *J. Syst. Archit.* 72 (2016) 93–107.
- [27] Y.C. Hu, M. Patel, D. Sabella, N. Sprecher, V. Young, Mobile edge computing: A key technology towards 5G, *ETSI White Pap.* 11 (11) (2015.) 1–16.
- [28] K. Bilal, O. Khalid, A. Erbad, S.U. Khan, Potentials, trends, and prospects in edge technologies: Fog, cloudlet, mobile edge, and micro data centers, *Comput. Netw.* 130 (2018.) 94–120.
- [29] D. Linthicum, Edge computing vs. fog computing: definitions and enterprise uses, 2018. Available online: <https://www.cisco.com/c/en/us/solutions/enterprise-networks/edge-computing.html> (Accessed 20 november 2018).
- [30] J. Zhang, B. Chen, Y. Zhao, X. Cheng, F. Hu, Data security and privacy-preserving in edge computing paradigm: survey and open issues, *IEEE Access* 6 (March) (2018) 18209–18237.
- [31] W. Yu, F. Liang, X. He, W. Grant, C. Lu, J. Lin, X. Yang, A survey on the edge computing for the internet of things, *IEEE Network* 6 (1) (2018) 6900–6919.
- [32] F. Ganz, D. Puschmann, P. Barnaghi, F. Carrez, A practical evaluation of information processing and abstraction techniques for the internet of things, *IEEE Internet Things J.* 2 (4) (2015.) 340–354.
- [33] M. Razzaque, M. Milojevic-Jevric, A. Palade, S. Clarke, Middleware for internet of things: a survey, *IEEE Internet Things J.* 3 (1) (2016) 70–95.
- [34] G. Premsankar, M. Di Francesco, T. Taleb, Edge computing for the internet of things: A Case study, *IEEE Internet Things J.* 5 (2) (2018) 1275–1284.
- [35] Q. Jing, A.V. Vasilakos, J. Wan, J. Lu, D. Qiu, Security of the internet of things: perspectives and challenges, *Wirel. Netw.* 20 (8) (2014.) 2481–2501.
- [36] A. Brogi, S. Forti, Qos-aware deployment of iot applications through the fog, *IEEE Internet Things J.* 4 (5) (2017) 1–8.
- [37] J. Lin, W. Yu, N. Zhang, X. Yang, H. Zhang, W. Zhao, A survey on internet of things: Architecture, enabling technologies, security and privacy, and applications, *IEEE Internet Things J.* 4 (5) (2017) 1125–1142.
- [38] M.A. Khan, K. Salah, Iot security: Review, blockchain solutions, and open challenges, *Future Gener. Comput. Syst.* 82 (2018) 395–411.
- [39] M. Banerjee, J. Lee, K.-K.R. Choo, A blockchain future for internet of things security: a position paper, *Digit. Commun. Networks* 4 (3) (2018) 149–160.
- [40] X. Li, P. Jiang, T. Chen, X. Luo, Q. Wen, A survey on the security of blockchain systems, *Future Gener. Comput. Syst.* (2017).
- [41] T.M. Fernández-Caramés, P. Fraga-Lamas, A review on the use of blockchain for the internet of things, *IEEE Access* 6 (2018) 32979–33001.
- [42] S. Raval, *Decentralized Applications: Harnessing Bitcoin's Blockchain Technology*, O'Reilly Media, Inc., 2016.
- [43] Z. Xiong, Y. Zhang, D. Niyato, P. Wang, Z. Han, When mobile blockchain meets edge computing, *IEEE Commun. Mag.* 56 (8) (2018) 33–39.
- [44] M. Isaja, J. Soldatos, V. Gezer, Combining edge computing and blockchains for flexibility and performance in industrial automation, *Int. Conf. Mobile Ubiquitous Comput. Syst. Serv. Technol. (UBICOMM)* (c) (2017.) 159–164.
- [45] F. Bonomi, R. Milito, J. Zhu, S. Addepalli, Fog computing and its role in the internet of things, in: *Proceedings of the First Edition of the MCC Workshop on Mobile Cloud Computing - MCC '12*, 2012., p. 13.
- [46] W. Shi, D. Schahram, The promise of edge computing, *Computer* 49 (0018) (2016) 78–81.
- [47] M. Moghaddam, M.N. Cadavid, C.R. Kenley, A.V. Deshmukh, Reference architectures for smart manufacturing: A critical review, *J. Manuf. Syst.* 49 (2018) 215–225.
- [48] U. Van Heesch, P. Avgeriou, R. Hilliard, A documentation framework for architecture decisions, *J. Syst. Softw.* 85 (2011) 795–820.
- [49] OpenFog Consortium, OpenFog Reference Architecture for Fog Computing, Technical Report, OpenFog Consortium, 2017. Available online: https://www.openfogconsortium.org/wp-content/uploads/OpenFog_Reference-Architecture_2_09_17-FINAL.pdf (Accessed 20 november 2018).
- [50] R. Faia, T. Pinto, Z. Vale, Dynamic Fuzzy clustering method for decision support in electricity markets negotiation, *ADCAIJ: Advances in Distributed Computing and Artificial Intelligence Journal* 5 (1) (2016) 23–35–35.
- [51] S. Nakamoto, Bitcoin open source implementation of p2p currency, *P2P Found.* 18 (2009).
- [52] E. Querol, J.A. Romero, A.M. Estruch, F. Romero, Norma iec-61499 para el control distribuido. Aplicación Al Cnc, *Actas XXXV Jornadas Autom.* (2014) 8.
- [53] A. Stanciu, Blockchain based distributed control system for edge computing, in: *Proceedings - 2017, 21st International Conference on Control Systems and Computer, CSCS 2017*, 2017, pp. 667–671.
- [54] Postscapes, IoT Standards & Protocols Guide - 2019 Comparisons on Network, Wireless Comms, Security, Industrial, 2019. Available online: <https://www.postscapes.com/internet-of-things-protocols/>. (Accessed 20 january 2019).
- [55] A. Ali, G.A. Shah, M.O. Farooq, U. Ghani, Technologies and challenges in developing machine-to-machine applications: A survey, *J. Netw. Comput. Appl.* 83 (2017) 124–139.
- [56] K. Cho, G. Park, W. Cho, J. Seo, K. Han, Performance analysis of device discovery of bluetooth low energy (BLE) networks, *Comput. Commun.* 81 (2016) 72–85.
- [57] F. Montori, L. Bedogni, M. Di Felice, L. Bononi, Machine-to-machine wireless communication technologies for the internet of things: Taxonomy, comparison and open issues, *Pervasive Mob. Comput.* 50 (2018) 56–81.

- [58] LoRa-Alliance, A technical overview of LoRa and Lo-RaWAN What is it?, Technical Report, 2015. Available online: https://www.tuv.com/media/corporate/products_1/electronic_components_and_lasers/TUeV_Rheinland_Overview_LoRa_and_LoRaWANtmp.pdf. (Accessed 20 november 2018).
- [59] M. Samaniego, R. Deters, Hosting virtual iot resources on edge-hosts with blockchain, in: Proceedings - 2016, 16th IEEE International Conference on Computer and Information Technology, CIT., IEEE, 2016, pp. 116–119.
- [60] National Institute of Standards and Technology, Department of Commerce, United State of America, Secure Hash Standard (SHS), Technical Report, Department of Commerce, United States of America, 2015. Available online: <https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.180-4.pdf>. (Accessed 20 november 2018).
- [61] G. Liang, S.R. Weller, F. Luo, J. Zhao, Z.Y. Dong, Distributed blockchain-based data protection framework for modern power systems against cyber attacks, IEEE Trans. Smart Grid 3053 (c) (2018) 1–12.
- [62] M. Pilkington, 11 blockchain technology: principles and applications, in: Research Handbook on Digital Transformations, Edward Elgar Publishing, 2016, p. 461.
- [63] R. Rivest, A. Shamir, L. Adleman, A method for obtaining digital signatures and public-key cryptosystems, M.I.T. Lab. Comput. Sci. Tech. Memo 82 (1977) 120–126.
- [64] W. Stallings, L. Brown, M.D. Bauer, A.K. Bhattacharjee, Computer security: principles and practice, Pearson Education, 2012.
- [65] BlockchainHub, What's a blockchain oracle?, 2019. Available online: <https://blockchainhub.net/blockchain-oracles/> (Accessed 20 january 2019).
- [66] J. Tang, Intelligent Mobile Projects with TensorFlow: Build 10+ Artificial Intelligence apps using TensorFlow Mobile and Lite for iOS, Android, and Raspberry Pi, Packt Publishing, 2018.
- [67] A. Das, S. Patterson, M. Wittie, Edgebench: Benchmarking edge computing platforms, in: 2018 IEEE/ACM International Conference on Utility and Cloud Computing Companion (UCC Companion), IEEE, 2018, pp. 175–180.
- [68] I. Sittón-Candanedo, S. Rodríguez, Pattern extraction for the design of predictive models in industry 4.0, in: International Conference on Practical Applications of Agents and Multi-Agent Systems, Springer, 2017, pp. 258–261.
- [69] A. González-Briones, P. Chamoso, H. Yoe, J.M. Corchado, Greenvmas: virtual organization based platform for heating greenhouses using waste energy from power plants, Sensors 18 (3) (2018) 861.
- [70] R.S. Alonso, D.I. Tapia, J. Bajo, O. García, J.F. de Paz, J.M. Corchado, Implementing a hardware-embedded reactive agents platform based on a service-oriented architecture over heterogeneous wireless sensor networks, Ad Hoc Networks 11 (1) (2013) 151–166.
- [71] A. González-Briones, P. Chamoso, S. Rodríguez, H. Yoe, J.M. Corchado, Reuse of waste energy from power plants in greenhouses through MAS-based architecture, Wirel. Commun. Mobile Comput. 2018 (2018).
- [72] J. Kaufman, A. Saxton, A. Rius, Relationships among temperature-humidity index with rectal, udder surface, and vaginal temperatures in lactating dairy cows experiencing heat stress, J. Dairy Sci. (2018).
- [73] J. Mankar, C. Darode, K. Trivedi, M. Kanoje, P. Shahare, Review of i2c protocol, Int. J. 2 (1) (2014) 474–479.



Ricardo S. Alonso is an experienced Engineer with a demonstrated history of working in the information technology and services industry. Skilled in Wireless Sensor Networks, Real-Time Locating Systems, Embedded Systems, Linux, Mobile Applications and Software Project Management. Strong engineering professional graduated from University of Valladolid (Telecommunications Engineer) and University of Salamanca (MSc in Intelligent Systems). He is currently working on his Ph.D. in Computer Engineering as a member of the BISITE research group.



Juan M. Corchado is Full Professor with Chair at the University of Salamanca. He earned a Ph.D. in Computer Sciences from the University of Salamanca and he also holds a Ph.D. in Artificial Intelligence from the University of the West of Scotland. He is Visiting Professor at Osaka Institute of Technology since January 2015, Visiting Professor at the University Teknologi Malaysia since January 2017 and a Member of the Advisory group on Online Terrorist Propaganda of the European Counter Terrorism Centre (EUROPOL).



Sara Rodríguez-González received a Ph.D. in Computer Science from the University of Salamanca in 2010. She pursued her studies of Ph.D. in this University. She obtained a Technical Engineering in Systems Computer Sciences degree in 2004, an Engineering in Computer Sciences degree in 2007 at the University of Salamanca. She is Associate Professor at the University of Salamanca and researcher at the BISITE research group (<http://bisite.usal.es>). She has participated as a co-author in papers published in recognized international conferences and symposiums.



Roberto Casado-Vara earned a degree in Mathematics from the University of Salamanca, a Master in Big Data, Visual Analytics at the International University of la Rioja. He has worked at Viewnext as a data scientist and Powercenter consultant for important clients in the pharmaceutical and public administration sectors.

He is currently working on his Ph.D. in Computer Engineering as a member of the BISITE research group. As a researcher, his interests are focused on Deep Learning, advanced mathematical models for intelligent robust and non-linear control and monitoring, blockchain and knowledge discovery data as well as other fields.



Inés Sittón-Candanedo earned a degree in Computer Systems Engineer from the University Latina of Panama and a Master in Technologies and Information Systems from the UNACHI — University of Chiriquí, Panama. She is currently working on her Ph.D. in Computer Engineering as a member of the BISITE research group. As a researcher, her interests are focused on Internet of Things, Industry 4.0, Edge Computing, blockchain and other fields.